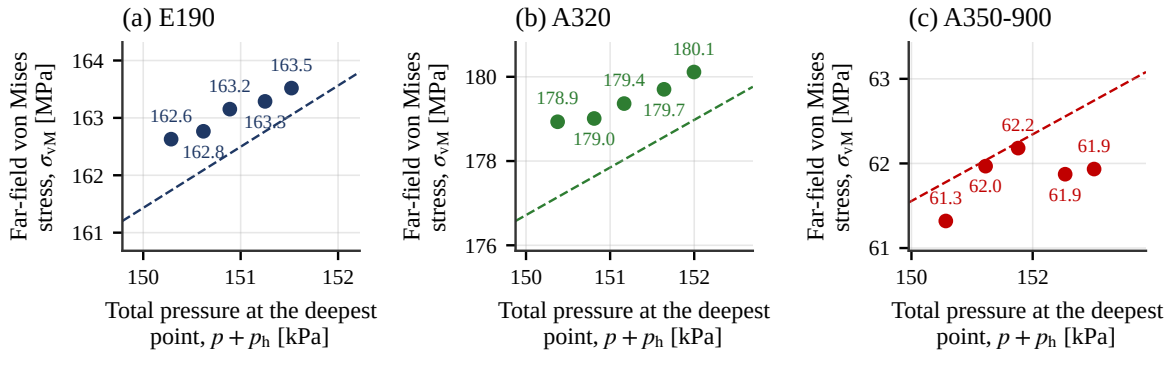
(a) E190
(b) A320
(c) A350-900
Far-field von Mises stress, σ_{vM} [MPa]
Total pressure at the deepest point, $p + p_h$ [kPa]
162.6
162.8
163.2
163.3
163.5
178.9
179.0
179.4
179.7
180.1
61.3
62.0
62.2
61.9
61.9
Finite element result at fill levels 0.10, 0.30, 0.50, 0.75 and 0.90
Line $a\,(p + p_h)\,R/t + b$ fitted to all twenty cases of each aircraft:
E190 a = 1.055, b = 1.5 MPa, A320 a = 1.039, b = 7.3 MPa, A350-900 a = 0.868, b = 2.4 MPa